Machine Learning
COMP4702/7703

COORDINATOR: A/PROF MARCUS GALLAGHER

SEMESTER 1, 2026

Welcome!

Today and this week:

- How the course will run this semester, ECP stuff, admin.
- Introduction
- Background

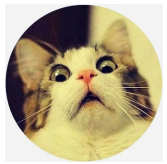
Course Staff

- A/Prof Marcus Gallagher
- https://www.linkedin.com/in/marcus-r-gallagher
- PhD (UQ), BCompSc, GradDipSc (UNE),

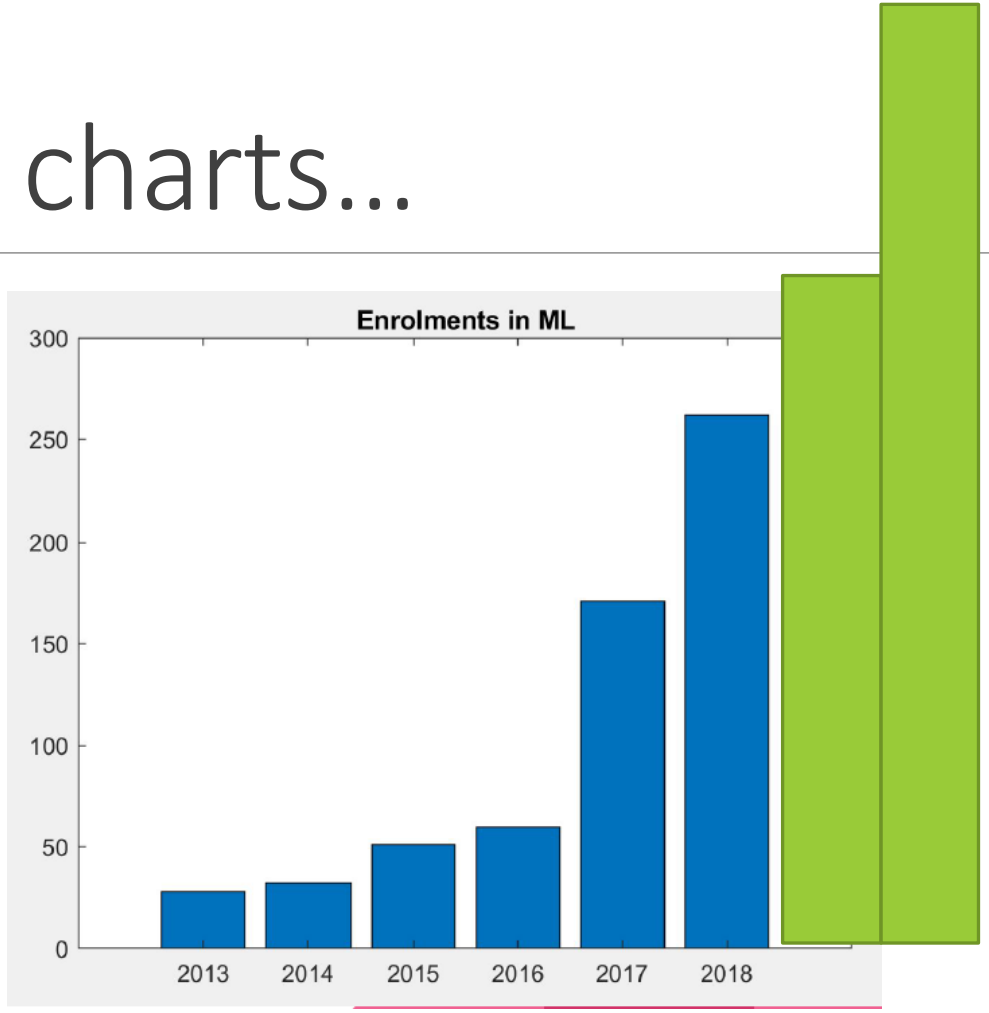
GradCert(HigherEd)(UQ)

- Teaching at UQ since 1999
- Research interests: machine learning, optimisation algorithms, understanding the nature of problems.
  - https://scholar.google.com/citations?user=Wcdm1ycAAAAJ&hl=en
- I am lucky to have a team of **awesome tutors** to help me with the course!

ML and AI are off the charts…



- The size of this course is unprecedented.
- What I care about MOST is that you have a good Learning experience in this course
  - But I am just one person and there are only so many hours in the day
- In order to maintain my sanity:
  - 90+% of questions resolved via Ed Discussion Board
  - Tutors will help me with admin wherever possible
  - Try all other options first **before** emailing me!
  - I won't send unnecessary emails – e.g. if the issue has already been resolved.
  - If it's a personal issue or needs course coordinator attention then please email me.



Generative AI (e.g. ChatGPT)

- LLMs (and other GenAI, ML) are surely going to disrupt many industries and aspects of our lives.
  - Education is an excellent example!
- They are a type of generative “AI” model. Other recent notable examples are text to image generation, text to music.
- UQ has a statement about using generative AI – course coordinators are given scope to do something they think sensible for their course.
- I think trying to “ban” LLMs is a waste of time.
- What happens in higher education?
  - You take courses (learning)
  - Uni makes you do assessment (did you learn something?)
  - Uni gives you a stamp to say “we think you learned something”
- How do we really know that you learned something?

Using GenAI in this course

- Sometimes, “old tech” is the best answer we have!
  - Identify-verified assessment (final exam, prac demos)
- Assignment:
  - You are free to use LLMs provided you acknowledge/cite your usage.
  - The rubric may include reflection on LLM results, context from class, etc.
- Use of LLMs in pracs is encouraged (for tutors as well as students!)
- I fully intend to have some fun with LLMs in the course!
  - But we will NOT be using them to assess your learning.
- HOWEVER:
  - **Don’t let LLMs rob you of what you can learn – this is a BIG concern**.

The Human Touch!

I assume you are in this course to learn.

Some things that help humans to learn?

- Cohort/social benefits
- Connecting learning with physical action
  - Health and wellbeing
- Connecting learning with emotions
  - Health and wellbeing
- Curiosity
- Feedback
- Engagement and reflection
- Carrots and sticks

The classes and assessment in this course have been increasingly guided by thinking about these things.

Vital Info for This Semester

- Ed Discussion Board will be an essential part of the course.
- Assessment and Classes (see course profiles for more info).
- Final exam (worth 60%, hurdles of 35% and 50%), assignment (22%).
  - Marking rubric will be different for the assignment for COMP4702 v's COMP7703.
- Lectures. Tied to assessment (participation, 18%) but everyone is still welcome to attend. Not "watching a show": it's a discussion around the course material for that week.
- Practical classes: mostly self-directed learning, completing the practical questions each week. Tutors are there to answer questions and help. Tied to assessment (demos, 18%).

How to get a “7” in this course?

- ✓ Engage! If you focus, contribute and put a lot of effort in, you will get a lot out.
- ✓ Attend class. Bring paper and pen (or maybe tablet)!
- ✓ Read the book each week (before or after the lecture).
- ✓ Do the pracs.
- ✓ Use Ed (engage, ask questions).
- ✓ Put in about 10 hrs per week of time for this course.
- ✓ Have some fun! ☺

What is Machine Learning?

- A subfield of Artificial Intelligence.
  - Goal: to create computer systems that are intelligent (strong), or fool you into thinking they are (weak).
  - One thing we associate with (natural) intelligent entities: adaptive behaviour, and/or the ability to learn.
  - So machine learning is about creating computer systems that can learn to perform tasks.

In other words, machine learning software automatically builds software to perform a task.

- On the basis of…
  - **DATA!**
- So machine learning is about engineering algorithms that learn from data to solve interesting problems.
- "…the study of computer algorithms capable of learning to improve their performance on a task on the basis of their own experience."
- This is sometimes called **inductive learning**.

Learning is an extremely broad concept, studied by psychologists, neuroscientists and others.

Machine learning restricts itself to a few well-defined classes of learning problems, that are still VERY general.

Furthermore, we concentrate mainly on problems that involve numerical data, or at least “structural” data

- e.g. nothing that involves language comprehension or “high-level” cognitive-type stuff!
- Our problems might be called sub-symbolic, or low-level…

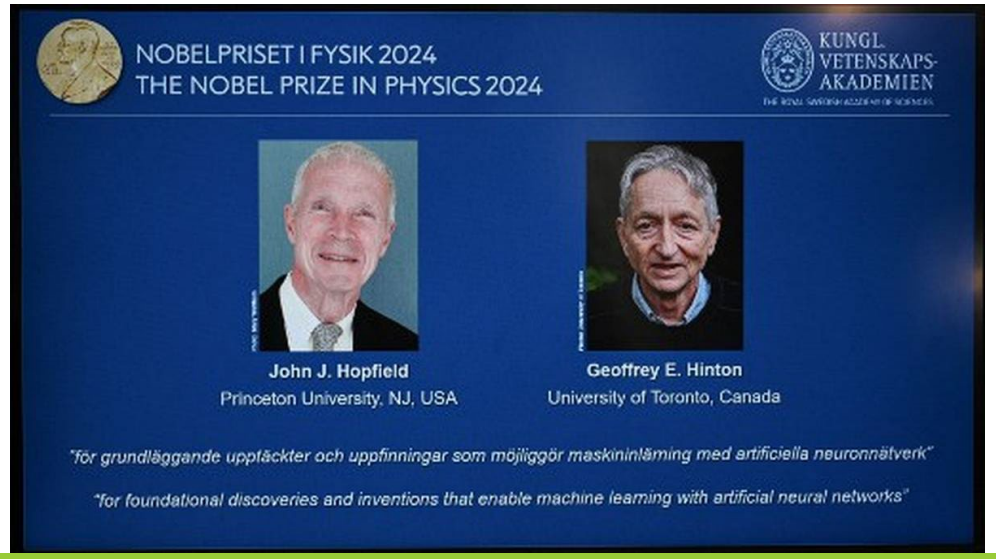
The Rise of the (Learning) Machines!

Machine Learning (incorrectly aka Artificial Intelligence) is one of the hottest topics on the planet.

The surge in interest and attention in the last few years has been fast, frantic and exciting.

AI has suddenly achieved incredible performance

- Enough to start serious discussions about “AGI”, existential threats, etc.



The Price of Fame and Fortune?

Currently HUGE money, interest and effort in AI

Scientific rigour is incredibly important but is now extremely weak in AI!

The objectives and applications of AI connect directly to issues of humanity such as ethics, bias, explainability, safety, trust, etc.

With critical mass, these factors can lead to pretty volatile situations and drama that computer scientists would not normally be involved in!